

KOUSTUBH PANDE

Bhopal, Madhya Pradesh, India

+91 91310 07294 | koustubhpande021@gmail.com | [linkedin/koustubh-pande](https://www.linkedin.com/in/koustubh-pande) | [github/wurtz-codez](https://github.com/wurtz-codez) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full Stack Developer and Computer Science undergraduate with hands-on experience across the MERN stack, cloud deployment, and AI-driven systems. Proficient in React.js, Next.js, Node.js, and Python, with practical experience in REST API design, authentication, containerized deployments, and CI/CD pipelines. Built and shipped production applications end-to-end as a founding engineer and freelance developer, including LLM-integrated tools using RAG and real-time voice AI. Comfortable owning a project from system design through deployment, with a strong focus on performance, scalability, and clean architecture.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering — CGPA: 8.4 / 10.0

September 2023 – Present

Bhopal, Madhya Pradesh

TECHNICAL SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, C++, Java
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS, HTML5, CSS3
- **Backend & APIs:** Node.js, Express.js, Flask, Django, REST APIs, Authentication (JWT, OAuth)
- **Databases:** MongoDB, PostgreSQL, MySQL
- **Cloud & Deployment:** AWS, Docker, Vercel, Render, Netlify, CI/CD
- **AI & Machine Learning:** RAG, LangChain, LLM APIs, transformers.js, Generative AI
- **Tools:** Git, GitHub, Cursor, Figma, Framer, Postman

EXPERIENCE

Founding Engineer / Co-founder — Singularity Works

Aug 2025 – Present

Remote

- Architected a scalable product architecture using a monorepo-based microservices approach with pnpm, Bun, and TurboRepo, integrating CI/CD pipelines and improving deployment workflows by 30%
- Developed 40+ reusable, component-driven UI modules in React.js, applying code splitting and lazy loading to cut load time 40–50% and achieve 90+ Lighthouse scores
- Planned and streamlined REST API architecture and integrations, implementing efficient API design patterns and Docker-containerized services for full CI/CD environment parity

Full Stack Developer — Freelancer

Dec 2024 – Jun 2025

Remote

- Built and launched a full-stack e-commerce platform for a jewelry startup on the MERN stack, driving 25% revenue growth and a 20% improvement in customer retention within the first quarter post-launch
- Executed a role-based admin dashboard with secure JWT-based RBAC authentication, enabling real-time inventory and order management while reducing manual workload by 40%
- Expanded a React and Redux-based gamified web app with an Appwrite backend, boosting engagement by 30%; delivered responsive frontends for a school organization and a designer portfolio using Next.js and Tailwind CSS

PROJECTS

Qode – AI-Powered Visual Codebase Exploration Tool

React, TypeScript, Graph RAG, KuzuDB (WASM), Tree-sitter, Sigma.js, LangChain, transformers.js | [Live Link](#)

- Engineered a client-side Graph RAG system, applying system design principles to transform codebases into knowledge graphs using KuzuDB and Tree-sitter, enabling context-aware LLM retrieval and reducing hallucination
- Constructed hybrid search using transformers.js with WebGPU acceleration, combining keyword matching with semantic embeddings for accurate, fully in-browser code retrieval
- Delivered a WebGL-powered visual graph explorer with Sigma.js and community detection algorithms (Leiden/Louvain), maintaining 60fps performance; configured GitHub Actions CI/CD with ruff, pyright, and pytest for automated linting, type-checking, and test coverage

VerQ – AI-Powered Interview Buddy

React, Node.js, MongoDB, Express, Gemini API, Deepgram API, RAG | [GitHub](#)

- Built a full-stack AI interview platform using the MERN stack, implementing RAG pipelines and real-time voice processing with Gemini and Deepgram APIs, enabling personalized feedback with sub-300ms latency
- Established a low-latency voice transcription pipeline with Deepgram streaming and Gemini prompt chaining; implemented secure JWT authentication and optimized performance for real-time processing

Bazaario – Multi-Vendor E-Commerce Platform

React.js, Node.js, Express.js, MongoDB, JWT, Chart.js | [GitHub](#)

- Architected a scalable multi-vendor marketplace with customizable storefronts; implemented JWT-based RBAC authentication and created REST APIs for secure and efficient data flow
- Engineered an interactive Chart.js analytics dashboard featuring pie and line chart visualizations, empowering vendors to monitor sales trends and make data-driven inventory decisions, increasing operational efficiency by 15%

CERTIFICATIONS

- Google Professional Certificate — [Link](#)

ACHIEVEMENTS & LEADERSHIP

- Winner, HACKBYTE 4.0 – National-level 36-hour hackathon, IIITDM Jabalpur — [Link](#)
- Winner, 48-hour offline hackathon by GSoC Innovators Club, VITB (powered by Unstop & InterviewBuddy™)
- Tech Team Lead, Mozilla Firefox Club